

Nonlinear Estimation

ECMT 676 (Econometric II) Lecture Notes

TAMU

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- Up until now, all estimators we have studied can be written as “closed form” functions of the data.
- For example, the OLS estimator is

$$\begin{aligned}\hat{\beta}_{OLS} &= \left(\sum_{i=1}^N \mathbf{x}_i' \mathbf{x}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{x}_i' y_i \right) \\ &= \beta + \left(N^{-1} \sum_{i=1}^N \mathbf{x}_i' \mathbf{x}_i \right)^{-1} \left(N^{-1} \sum_{i=1}^N \mathbf{x}_i' u_i \right).\end{aligned}$$

- Estimators in closed form do not cover all cases of interest, particularly when we turn to nonlinear models.
- Even if the underlying model is linear, special asymptotic methods are sometimes needed for certain estimators.

- Suppose that

$$\text{Med}(y|\mathbf{x}) = \mathbf{x}\boldsymbol{\beta} = \beta_1 + \beta_2 x_2 + \dots + \beta_K x_K$$

is the conditional median of y given $\mathbf{x}$.

- Without additional assumptions, OLS does not consistently estimate the β_j . But *least absolute deviations* (LAD) does.
- The LAD estimator solves

$$\min_{\mathbf{b} \in \mathbb{R}^K} \sum_{i=1}^N |y_i - \mathbf{x}_i \mathbf{b}|,$$

that is, it minimizes the sum of absolute deviations (or residuals).

- For large-sample analysis, a key point is that the LAD estimate cannot generally be written in closed form.

- Suppose that a solution to the problem does exist; call it $\hat{\beta}_{LAD}$.
- Then we know

$$\hat{\beta}_{LAD} = \mathbf{g}_N(\mathbf{x}_1, y_1, \mathbf{x}_2, y_2, \dots, \mathbf{x}_N, y_N)$$

for some function $\mathbf{g}_N(\cdot)$. The problem is we do not know $\mathbf{g}_N(\cdot)$.

- Question: If we do not know $\mathbf{g}_N(\cdot)$, how do we study the large-sample (asymptotic) properties of $\hat{\beta}_{LAD}$?
- Answer: Indirectly, through the properties of the objective function.

- In particular, for each (dummy argument) $\mathbf{b}$,

$$N^{-1} \sum_{i=1}^N |y_i - \mathbf{x}_i \mathbf{b}|$$

is an average of independent, identically distributed random variables, $q_i(\mathbf{b}) \equiv |y_i - \mathbf{x}_i \mathbf{b}|$, and so we can apply the law of large numbers if $E[q_i(\mathbf{b})] < \infty$.

- The same issue arises for nonlinear models.
- Suppose for $y \geq 0$ we specify an exponential conditional mean model:

$$E(y|\mathbf{x}) = \exp(\mathbf{x}\boldsymbol{\beta}) = \exp(\beta_1 + \beta_2 x_2 + \dots + \beta_K x_K).$$

- Without further assumptions, we cannot “linearize” the model by using $\log(y)$ as the dependent variable and then apply OLS. [If $P(y = 0) > 0$, $\log(y)$ is not even defined.]

- Instead, we can directly estimate β by *nonlinear least squares* (NLS):

$$\min_{\mathbf{b} \in \mathbb{R}^K} N^{-1} \sum_{i=1}^N [y_i - \exp(\mathbf{x}_i \mathbf{b})]^2.$$

- As in the case of LAD, we cannot present the solution in closed form. But the estimator minimizes a function that is an average of i.i.d. random functions of $\mathbf{b}$.
- For our purposes, “nonlinear” means any situation where an estimator cannot be obtained in closed form. This requires a new set of tools for asymptotic analysis.

Consistency of M-Estimators

- We first cover a class of estimation problems estimators known as *M-estimation*. (The “M” refers, for us, to “minimization.” Originally, M-estimators we defined as maximization problems.)
- We will carry along the example of nonlinear least squares for a general regression function.
- Because we will require a separate notation for the value of the parameters describing the population, and the set of candidates for those values, we introduce a new convention.

- Consider a linear regression model where we know the population parameter values, say

$$E(y|x, z) = 3.26 + 0.75 x - 1.84 z.$$

This population regression is a particular version of the *model*

$$m(x, z) = \theta_1 + \theta_2 x + \theta_3 z,$$

for $E(y|x, z)$, where each θ_j can range across all real numbers.

- It is helpful to let $(\theta_1, \theta_2, \theta_3)$ denote a generic candidate for the vector of population parameters.
- The actual population parameters are denoted θ_{o1}, θ_{o2} , and θ_{o3} . That is, $\theta_{o1} = 3.26$, $\theta_{o2} = 0.75$, and $\theta_{o3} = -1.84$.

- In practice, we do not know the vector of values, θ_o , actually describing the population.
- In order to state assumptions in general nonlinear contexts, we need to distinguish between θ_o and a generic vector, θ .
- For NLS, we specify a model for the conditional mean, $E(y|\mathbf{x})$, where y is a scalar response and $\mathbf{x}$ is a vector.
- We focus on *parametric* models, which means the function is known up to the unknown parameters.
- Let $m(\mathbf{x}, \theta)$ represent model for all $\mathbf{x} \in \mathcal{X}$ and $\theta \in \Theta \subset \mathbb{R}^P$, where P is a positive integer.

- So θ is a $P \times 1$ vector. The *parameter space* Θ is the set of all parameters values that are candidates for the population value.
- As an example, $m(\mathbf{x}, \theta) = \exp(\mathbf{x}\theta) = \exp(\theta_1 + \theta_2 x_2 + \dots + \theta_K x_K)$ where $\mathbf{x} = (1, x_2, \dots, x_K)$ contains unity for convenience.
- The parameter space is probably $\Theta = \mathbb{R}^K$ because it is unlikely we would restrict it ahead of time.
- If $0 \leq y \leq 1$ – sometimes called a *fractional response* – a sensible model is

$$m(\mathbf{x}, \theta) = \frac{\exp(\theta_1 + \theta_2 x_2 + \dots + \theta_K x_K)}{1 + \exp(\theta_1 + \theta_2 x_2 + \dots + \theta_K x_K)} = \frac{\exp(\mathbf{x}\theta)}{1 + \exp(\mathbf{x}\theta)} \equiv \Lambda(\mathbf{x}\theta).$$

- For much of our development, we assume the model *correctly specified* for the conditional mean.

ASSUMPTION NLS.1: For some $\theta_o \in \Theta$,

$$E(y|\mathbf{x}) = m(\mathbf{x}, \theta_o), \mathbf{x} \in \mathcal{X}. \quad \square$$

- θ_o is called the “true value of the parameters.”
- For some purposes, it is useful to write the equation in error form:

$$\begin{aligned} y &= m(\mathbf{x}, \theta_o) + u \\ E(u|\mathbf{x}) &= 0, \end{aligned}$$

where the zero conditional mean holds by construction.

- Generally, other features of $D(u|\mathbf{x})$ are unrestricted.
- We should avoid thinking that u must be independent of $\mathbf{x}$. We should not even think homoskedasticity, $\text{Var}(u|\mathbf{x}) = \text{Var}(u)$, holds in general.

- how should we estimate θ_o ?
- Recall an important feature of the conditional mean. Assume $E(y^2) < \infty$ and let $E(y|\mathbf{x}) = \mu_o(\mathbf{x})$. Then among all functions $\mu(\mathbf{x})$ with $E[\mu(\mathbf{x})^2] < \infty$,

$$E\{[y - \mu_o(\mathbf{x})]^2\} \leq E\{[y - \mu(\mathbf{x})]^2\}.$$

That is, the conditional mean is the minimum mean square predictor of y .

- It follows that if $m(\mathbf{x}, \theta)$ is a correctly specified model of $E(y|\mathbf{x})$, then

$$E\{[y - m(\mathbf{x}, \theta_o)]^2\} \leq E\{[y - m(\mathbf{x}, \theta)]^2\}, \text{ all } \theta \in \Theta.$$

- A direct proof is constructive. Write $y = m(\mathbf{x}, \theta_o) + u$ and plug in:

$$\begin{aligned} [y - m(\mathbf{x}, \theta)]^2 &= [m(\mathbf{x}, \theta_o) + u - m(\mathbf{x}, \theta)]^2 \\ &= u^2 + 2[m(\mathbf{x}, \theta_o) - m(\mathbf{x}, \theta)]u + [m(\mathbf{x}, \theta_o) - m(\mathbf{x}, \theta)]^2 \end{aligned}$$

- Now take expectations:

$$\begin{aligned} E\{[y - m(\mathbf{x}, \theta)]^2\} &= E(u^2) + E\{2[m(\mathbf{x}, \theta_o) - m(\mathbf{x}, \theta)]u\} \\ &\quad + E\{[m(\mathbf{x}, \theta_o) - m(\mathbf{x}, \theta)]^2\} \\ &= E(u^2) + E\{[m(\mathbf{x}, \theta_o) - m(\mathbf{x}, \theta)]^2\} \end{aligned}$$

because $E(u|\mathbf{x}) = 0$.

- The proof is completed by noting that $E(u^2)$ does not depend on θ and $E\{[m(\mathbf{x}, \theta_o) - m(\mathbf{x}, \theta)]^2\}$ is smallest when $\theta = \theta_o$.

- We have shown that

$$\theta_o = \operatorname{argmin}_{\theta \in \Theta} E\{[y - m(\mathbf{x}, \theta)]^2\}.$$

- In other words, θ_o solves a population minimization problem.
- We will have to impose extra assumptions to ensure the solution is unique.
- The *analogy principle* says to solve the sample analog of the population problem, which leads to nonlinear least squares:

$$\min_{\theta \in \Theta} N^{-1} \sum_{i=1}^N [y_i - m(\mathbf{x}_i, \theta)]^2.$$

- The M-estimation principle generalizes this reasoning. We assume that $\theta_o \in \Theta$ *uniquely* solves

$$\min_{\theta \in \Theta} E[q(\mathbf{w}, \theta)],$$

where $q : \mathcal{W} \times \Theta \rightarrow \mathbb{R}$ is a real valued function of an observable vector $\mathbf{w}$ and the parameter vector θ .

- An M-estimator of θ_o solves the sample analog,

$$\min_{\theta \in \Theta} N^{-1} \sum_{i=1}^N q(\mathbf{w}_i, \theta).$$

- Does it seem reasonable that a solution, say $\hat{\theta} = \mathbf{g}_N(\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_N)$, is consistent for θ_o ?

- By the law of large numbers, for each θ ,

$$N^{-1} \sum_{i=1}^N q(\mathbf{w}_i, \theta) \xrightarrow{P} E[q(\mathbf{w}, \theta)]$$

$\hat{\theta}$ minimizes θ_o minimizes
(sample average) (population average)

So $\hat{\theta} \xrightarrow{P} \theta_o$ seems reasonable.

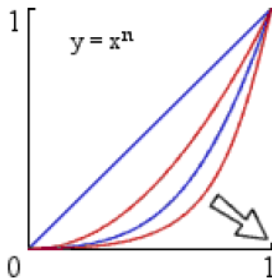
- But pointwise convergence of the sample objective function is not sufficient for consistency.
- A sufficient condition is *uniform convergence in probability*:

$$\max_{\theta \in \Theta} \left| N^{-1} \sum_{i=1}^N q(\mathbf{w}_i, \theta) - E[q(\mathbf{w}, \theta)] \right| \xrightarrow{p} 0$$

- Means that we can bound the distance between $N^{-1} \sum_{i=1}^N q(\mathbf{w}_i, \theta)$ and its expected value by something that does not depend on θ .

- In “regular” cases, the pointwise law of large numbers translates into the *uniform law of large numbers*.
- Sufficient conditions are
 1. $q(\mathbf{w}, \cdot)$ is continuous on Θ .
 2. Θ is closed and bounded (compact).
 3. $|q(\mathbf{w}, \theta)| \leq b(\mathbf{w})$ for some function $b(\mathbf{w})$ with $E[b(\mathbf{w})] < \infty$.
- For NLS, if $0 \leq y \leq 1$, then $|m(x, \theta)| \leq 1$.

Examples of uniform convergence.

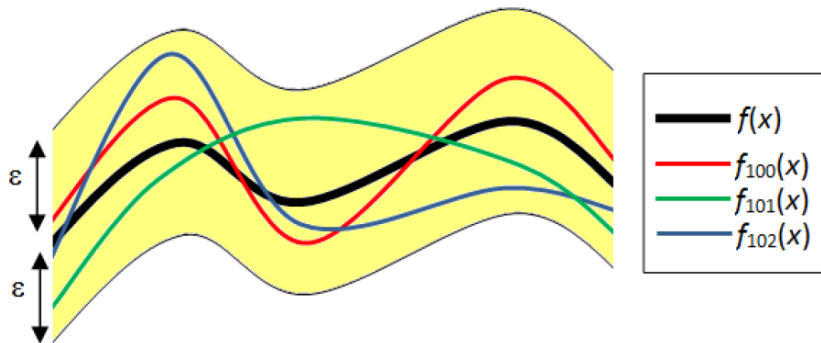


$$f_n(x) = x^n, x \in [0, 1]$$

$$f(x) = \begin{cases} 0, & \text{if } x \in [0, 1) \\ 1, & \text{if } x = 1 \end{cases}$$

$f_n(x) \rightarrow f(x)$, $x \in [0, 1]$, pointwise but *not* uniformly

$f_n(x) \rightarrow f(x)$, $x \in [0, 0.5]$, pointwise and uniformly



- Other than “regularity conditions” – continuity of $q(\mathbf{w}, \cdot)$ and finite moments – the key consistency assumption is identification.
- Namely, θ_o is the *unique* solution to the population problem.

EXAMPLE: Suppose $x > 0$ is a scalar, $y \geq 0$, and $m(x, \theta_1, \theta_2, \theta_3) = \theta_1 + \theta_2 x^{\theta_3}$, where the parameter spaces is $\Theta = [0, \infty) \times [0, \infty) \times \mathbb{R}$.

If $\theta_{o2} = 0$, so that $E(y|x) = E(y) = \theta_{o1}$, then any θ of the form $(\theta_{o1}, 0, \theta_3)'$ minimizes the mean squared error. What would NLS estimate for θ_{o3} ?

- We need an identification assumption for NLS that rules out the previous situation.

ASSUMPTION NLS.2: $E\{[m(\mathbf{x}, \theta_o) - m(\mathbf{x}, \theta)]^2\} > 0$ for all $\theta \in \Theta$, $\theta \neq \theta_o$. $\square$

- Assumption NLS.2 plays the role of a rank condition.
- In the linear case, $m(\mathbf{x}, \boldsymbol{\theta}) = \mathbf{x}\boldsymbol{\theta}$, and then

$$\begin{aligned} [m(\mathbf{x}, \boldsymbol{\theta}_o) - m(\mathbf{x}, \boldsymbol{\theta})]^2 &= [\mathbf{x}(\boldsymbol{\theta}_o - \boldsymbol{\theta})]^2 = (\boldsymbol{\theta}_o - \boldsymbol{\theta})' \mathbf{x}' \mathbf{x} (\boldsymbol{\theta}_o - \boldsymbol{\theta}) \\ E\{[m(\mathbf{x}, \boldsymbol{\theta}_o) - m(\mathbf{x}, \boldsymbol{\theta})]^2\} &= (\boldsymbol{\theta}_o - \boldsymbol{\theta})' E(\mathbf{x}' \mathbf{x}) (\boldsymbol{\theta}_o - \boldsymbol{\theta}) \end{aligned}$$

- For the last expression to be positive for all $\boldsymbol{\theta} \neq \boldsymbol{\theta}_o$, we need $E(\mathbf{x}' \mathbf{x})$ to have full rank K , which is exactly Assumption OLS.2.

- Theorem 12.2 contains a formal consistency result for general M-estimators.
- A practically important “regularity condition” is continuity of $q(\mathbf{w}, \cdot)$ over the parameter space. It can be relaxed to “continuity with probability one.”
- When $q(\mathbf{w}, \cdot)$ is continuous on Θ and Θ is compact, there is always a solution to

$$\min_{\theta \in \Theta} \sum_{i=1}^N q(\mathbf{w}_i, \theta).$$

- The solution to the sample problem need not be unique even though the solution to the population problem is unique.
- The LAD estimator is a good example where multiple solutions are possible even without perfect collinearity among the $\mathbf{x}_i$.

- Lemma 12.1 is a very useful result: Under finite moment conditions, if $r(\mathbf{w}, \cdot)$ is continuous on Θ ,

$$N^{-1} \sum_{i=1}^N r(\mathbf{w}_i, \theta) \xrightarrow{p} E[r(\mathbf{w}, \theta)], \text{ all } \theta \in \Theta,$$

and $\hat{\theta} \xrightarrow{p} \theta_o$, then

$$N^{-1} \sum_{i=1}^N r(\mathbf{w}_i, \hat{\theta}) \xrightarrow{p} E[r(\mathbf{w}, \theta_o)]$$

[The conditions ensure the uniform WLLN holds for $r(\mathbf{w}_i, \theta)$.]

- This result is useful for estimating average partial effects (which we will cover later) as well as asymptotic variance matrices.

Asymptotic Distribution

- We showed how consistency of M-estimators can be established without having closed form solutions. Now we turn to the question of approximating the sampling distribution of $\hat{\theta}$.
- We now add some smoothness assumptions. In particular, assume

$q(\mathbf{w}, \cdot)$ is twice continuously differentiable on $\text{int}(\Theta)$.

- Further, assume θ_o is in the interior of the parameter space:

$$\theta_o \in \text{int}(\Theta).$$

- Finite moment conditions are used, too.

- The gradient of $q(\mathbf{w}, \boldsymbol{\theta})$, defined on $\text{int}(\Theta)$, is the $1 \times P$ row vector

$$\nabla_{\boldsymbol{\theta}} q(\mathbf{w}, \boldsymbol{\theta}) = \left(\frac{\partial q(\mathbf{w}, \boldsymbol{\theta})}{\partial \theta_1} \quad \frac{\partial q(\mathbf{w}, \boldsymbol{\theta})}{\partial \theta_2} \quad \dots \quad \frac{\partial q(\mathbf{w}, \boldsymbol{\theta})}{\partial \theta_P} \right).$$

The *score* is the transpose of the gradient:

$$\mathbf{s}(\mathbf{w}, \boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} q(\mathbf{w}, \boldsymbol{\theta})'.$$

- Now, because $\boldsymbol{\theta}_o$ is in the interior of Θ and $\hat{\boldsymbol{\theta}} \xrightarrow{P} \boldsymbol{\theta}_o$, we know $\hat{\boldsymbol{\theta}} \in \text{int}(\Theta)$ with probability approaching one. We will ignore the qualifier here.

- Because $\hat{\theta}$ minimizes the sample objective function and is an interior solution, $\hat{\theta}$ solves

$$\sum_{i=1}^N \mathbf{s}(\mathbf{w}_i, \hat{\theta}) = \mathbf{0},$$

a set of P equations in P unknowns.

- By the mean value theorem (for each element of the score),

$$\sum_{i=1}^N s_m(\mathbf{w}_i, \hat{\theta}) = \sum_{i=1}^N s_m(\mathbf{w}_i, \theta_o) + \left(\sum_{i=1}^N \nabla_{\theta} s_m(\mathbf{w}_i, \ddot{\theta}_m) \right) (\hat{\theta} - \theta_o)$$

where $\ddot{\theta}_m$ is on the line segment between $\hat{\theta}$ and θ_o for $m = 1, \dots, P$.

- Therefore, $\ddot{\theta}_m \xrightarrow{P} \theta_o$. (In effect, $\ddot{\theta}_m$ is “trapped” between $\hat{\theta}$ and θ_o .)

- Stack all P elements to get

$$\sum_{i=1}^N \mathbf{s}(\mathbf{w}_i, \hat{\boldsymbol{\theta}}) = \sum_{i=1}^N \mathbf{s}(\mathbf{w}_i, \boldsymbol{\theta}_o) + \left(\sum_{i=1}^N \ddot{H}_i \right) (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o),$$

where $\ddot{H}_i$ is the $P \times P$ Hessian of $q(\mathbf{w}, \boldsymbol{\theta})$ – also the Jacobian of $\mathbf{s}(\mathbf{w}, \boldsymbol{\theta})$ – but with rows evaluated at generally different mean values.

- We will need the Hessian evaluated at a generic parameter vector:

$$\mathbf{H}(\mathbf{w}, \boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}}^2 q(\mathbf{w}, \boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \mathbf{s}(\mathbf{w}, \boldsymbol{\theta}) = \frac{\partial^2 q(\mathbf{w}, \boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}$$

- We will need the uniform convergence of $N^{-1} \sum_{i=1}^N \mathbf{H}(\mathbf{w}_i, \boldsymbol{\theta})$ over $\boldsymbol{\theta}$.

- Back to the score representation. Because $\hat{\boldsymbol{\theta}}$ solves the FOC,

$$\mathbf{0} = N^{-1/2} \sum_{i=1}^N \mathbf{s}(\mathbf{w}_i, \boldsymbol{\theta}_o) + \left(N^{-1} \sum_{i=1}^N \ddot{H}_i \right) \sqrt{N}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o).$$

- Because each $\ddot{\theta}_m \xrightarrow{p} \boldsymbol{\theta}_o$, $N^{-1} \sum_{i=1}^N \ddot{H}_i \xrightarrow{p} E[\mathbf{H}(\mathbf{w}, \boldsymbol{\theta}_o)] \equiv \mathbf{A}(\boldsymbol{\theta}_o) \equiv \mathbf{A}_o$ by Lemma 12.1 (here we need uniform convergence of $N^{-1} \sum_{i=1}^N \mathbf{H}(\mathbf{w}_i, \boldsymbol{\theta})$ over $\boldsymbol{\theta}$.)
- An assumption related to identification is that

$\mathbf{A}_o$ is positive definite

- Can show that $N^{-1} \sum_{i=1}^N \ddot{H}_i$ is nonsingular w.p.a.1. because it is getting “close” to $\mathbf{A}_o$, which is nonsingular.

- It follows that, w.p.a.1.,

$$\sqrt{N}(\hat{\theta} - \theta_o) = \left(N^{-1} \sum_{i=1}^N \ddot{H}_i \right)^{-1} \left[-N^{-1/2} \sum_{i=1}^N \mathbf{s}(\mathbf{w}_i, \theta_o) \right]$$

- Further, very generally the score has zero mean when evaluated at θ_o :

$$E[\mathbf{s}(\mathbf{w}, \theta_o)] = \mathbf{0}.$$

(Important: $E[\mathbf{s}(\mathbf{w}, \theta)] \neq \mathbf{0}$ for $\theta \neq \theta_o$.)

- Can show $E[\mathbf{s}(\mathbf{w}, \theta_o)] = \mathbf{0}$ if the derivative and expected value can be interchanged.
- By FOC in the population,

$$\nabla_{\theta} E[q(\mathbf{w}, \theta_o)] = \mathbf{0},$$

and so $E[\mathbf{s}(\mathbf{w}, \theta_o)] = \mathbf{0}$ if ∇_{θ} can pass through the expected value.

- Similar argument can be used to show $\mathbf{A}_o$ is positive definite.
- Because θ_o uniquely solves the population minimization problem,

$$\nabla_{\theta}^2 E[q(\mathbf{w}, \theta_o)]$$

is positive definite.

- Now interchange the partial derivatives with respect to θ and the expected value and we get $\mathbf{A}_o$.

- In other cases, such as NLS, can show directly the score has zero expected value at $\theta = \theta_o$.
- Why is $E[\mathbf{s}(\mathbf{w}, \theta_o)] = \mathbf{0}$ important?
- Because then, by the central limit theorem,

$$N^{-1/2} \sum_{i=1}^N \mathbf{s}(\mathbf{w}_i, \theta_o) \xrightarrow{d} \text{Normal}(\mathbf{0}, \mathbf{B}_o)$$

$$\mathbf{B}_o = \text{Var}[\mathbf{s}(\mathbf{w}_i, \theta_o)] = E[\mathbf{s}(\mathbf{w}_i, \theta_o)\mathbf{s}(\mathbf{w}_i, \theta_o)'].$$

- In particular, $N^{-1/2} \sum_{i=1}^N \mathbf{s}(\mathbf{w}_i, \theta_o) = O_p(1)$.

- Now,

$$\sqrt{N}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) = \mathbf{A}_o^{-1} \left[-N^{-1/2} \sum_{i=1}^N \mathbf{s}(\mathbf{w}_i, \boldsymbol{\theta}_o) \right] + o_p(1).$$

- If we define $\mathbf{r}_i(\boldsymbol{\theta}_o) = -\mathbf{A}_o^{-1} \mathbf{s}(\mathbf{w}_i, \boldsymbol{\theta}_o)$, then we can write

$$\sqrt{N}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) = N^{-1/2} \sum_{i=1}^N \mathbf{r}_i(\boldsymbol{\theta}_o) + o_p(1),$$

which is called the *influence function representation*.

- By the asymptotic normality of $N^{-1/2} \sum_{i=1}^N \mathbf{s}(\mathbf{w}_i, \boldsymbol{\theta}_o)$, we have

$$\sqrt{N}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) \xrightarrow{d} \text{Normal}(\mathbf{0}, \mathbf{A}_o^{-1} \mathbf{B}_o \mathbf{A}_o^{-1}).$$

- Generally, the asymptotic variance depends on the expected value of the Hessian and the variance of the score (both evaluated at $\boldsymbol{\theta}_o$).

- The expression for $Avar[\sqrt{N}(\hat{\theta} - \theta_o)]$ is of the “sandwich” form.
- We write

$$Avar(\hat{\theta}) = \mathbf{A}_o^{-1} \mathbf{B}_o \mathbf{A}_o^{-1} / N,$$

so that $\mathbf{A}_o^{-1} \mathbf{B}_o \mathbf{A}_o^{-1} / N$ is intended to approximate the actual sampling variation in $\hat{\theta}$ for a give sample size, N .

- Note that, as with simple estimators such as sample averages and OLS, $Avar(\hat{\theta})$ is of order $1/N$.

- The previous approach assumes $q(\mathbf{w}, \cdot)$ is twice continuously differentiable with the expected Hessian, $E[\mathbf{H}(\mathbf{w}_i, \theta_o)]$, positive definite.
- These assumptions rule out some important problems, such as LAD (and quantile regression more generally).
- The nondifferentiability of the absolute value function at zero is actually not the problem. It is the fact that the Hessian is identically zero.
- A different approach to asymptotic normality is needed for quantile regression; the estimators are still $\sqrt{N}$ -asymptotically normal.

Estimating the Asymptotic Variance

- For inference we need to consistently estimate $Avar[\sqrt{N}(\hat{\theta} - \theta_o)] = \mathbf{A}_o^{-1} \mathbf{B}_o \mathbf{A}_o^{-1}$. In other words, we need to consistently estimate $\mathbf{A}_o$ and $\mathbf{B}_o$.
- There are sometimes several different ways to estimate $\mathbf{A}_o$. An estimator that is always available is

$$N^{-1} \sum_{i=1}^N \mathbf{H}(\mathbf{w}_i, \hat{\theta}) = N^{-1} \sum_{i=1}^N \mathbf{H}_i(\hat{\theta}),$$

the average across units of the Hessians evaluated at the estimates.

- When $\mathbf{w}_i$ partitions as $(\mathbf{x}_i, \mathbf{y}_i)$, and we are correctly modeling a feature of $D(\mathbf{y}_i|\mathbf{x}_i)$, we can sometimes find an analytical formula for

$$\mathbf{A}(\mathbf{x}_i, \theta_o) \equiv E[\mathbf{H}(\mathbf{w}_i, \theta_o)|\mathbf{x}_i].$$

- By iterated expectations, $\mathbf{A}_o = E[\mathbf{A}(\mathbf{x}_i, \theta_o)]$.
- Thus, another consistent estimator of $\mathbf{A}_o$ is

$$N^{-1} \sum_{i=1}^N \mathbf{A}(\mathbf{x}_i, \hat{\theta}) = N^{-1} \sum_{i=1}^N \hat{\mathbf{A}}_i.$$

- This is the estimator based on the “expected Hessian,” although it is really the “conditional expected Hessian.”
- It is rarely possible to find the unconditional expected value of $\mathbf{H}(\mathbf{w}_i, \theta_o)$ when there are conditioning variables $\mathbf{x}_i$ because we do not usually model $D(\mathbf{x}_i)$.

- A natural, consistent estimator of $\mathbf{B}_o = E[\mathbf{s}(\mathbf{w}_i, \boldsymbol{\theta}_o)\mathbf{s}(\mathbf{w}_i, \boldsymbol{\theta}_o)']$ is

$$\hat{\mathbf{B}} = N^{-1} \sum_{i=1}^N \mathbf{s}(\mathbf{w}_i, \hat{\boldsymbol{\theta}})\mathbf{s}(\mathbf{w}_i, \hat{\boldsymbol{\theta}})' = N^{-1} \sum_{i=1}^N \mathbf{s}_i(\hat{\boldsymbol{\theta}})\mathbf{s}_i(\hat{\boldsymbol{\theta}})' = N^{-1} \sum_{i=1}^N \mathbf{s}_i \mathbf{s}_i'$$

- $\hat{\mathbf{B}}$ is called the *outer product of the score* estimator.

- Therefore,

$$\widehat{Avar}[\sqrt{N}(\hat{\theta} - \theta_o)] = \hat{\mathbf{A}}^{-1} \hat{\mathbf{B}} \hat{\mathbf{A}}^{-1} \xrightarrow{p} \mathbf{A}_o^{-1} \mathbf{B}_o \mathbf{A}_o^{-1}.$$

- As always, we define

$$\widehat{Avar}(\hat{\theta}) = \widehat{Avar}[\sqrt{N}(\hat{\theta} - \theta_o)] / N.$$

- Therefore,

$$\begin{aligned} \widehat{Avar}(\hat{\theta}) &= N^{-1} \left(N^{-1} \sum_{i=1}^N \hat{\mathbf{H}}_i \right)^{-1} \left(N^{-1} \sum_{i=1}^N \mathbf{s}_i \mathbf{s}_i' \right) \left(N^{-1} \sum_{i=1}^N \hat{\mathbf{H}}_i \right)^{-1} \\ &= \left(\sum_{i=1}^N \hat{\mathbf{H}}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{s}_i \mathbf{s}_i' \right) \left(\sum_{i=1}^N \hat{\mathbf{H}}_i \right)^{-1} \end{aligned}$$

- As with all other procedures, the divisions by N disappear in $\widehat{Avar}(\hat{\theta})$.

- If we can compute $\mathbf{A}(\mathbf{x}_i, \theta_o) = E[\mathbf{H}(\mathbf{w}_i, \theta_o) | \mathbf{x}_i]$ then we can use

$$\widehat{Avar}(\hat{\theta}) = \left(\sum_{i=1}^N \hat{\mathbf{A}}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{s}_i \mathbf{s}_i' \right) \left(\sum_{i=1}^N \hat{\mathbf{A}}_i \right)^{-1}.$$

- When the inverses exist, both estimators are always at least positive semidefinite, and usually positive definite unless the underlying model is poorly specified.

Nonlinear Least Squares

- Write $q(\mathbf{w}, \boldsymbol{\theta}) = [y - m(\mathbf{x}, \boldsymbol{\theta})]^2 / 2$ where $m(\mathbf{x}, \cdot)$ is twice continuously differentiable on $\text{int}(\Theta)$. Then

$$\begin{aligned}\nabla_{\boldsymbol{\theta}} q(\mathbf{w}, \boldsymbol{\theta}) &= -\nabla_{\boldsymbol{\theta}} m(\mathbf{x}, \boldsymbol{\theta}) [y - m(\mathbf{x}, \boldsymbol{\theta})] \\ \mathbf{s}(\mathbf{w}, \boldsymbol{\theta}) &= -\nabla_{\boldsymbol{\theta}} m(\mathbf{x}, \boldsymbol{\theta})' [y - m(\mathbf{x}, \boldsymbol{\theta})],\end{aligned}$$

- The NLS estimator satisfies

$$\sum_{i=1}^N \nabla_{\boldsymbol{\theta}} m(\mathbf{x}_i, \hat{\boldsymbol{\theta}})' [y_i - m(\mathbf{x}_i, \hat{\boldsymbol{\theta}})] = \mathbf{0}.$$

- In the linear case, $m(\mathbf{x}_i, \hat{\boldsymbol{\theta}}) = \mathbf{x}_i \hat{\boldsymbol{\theta}}$ and $\nabla_{\boldsymbol{\theta}} m(\mathbf{x}_i, \hat{\boldsymbol{\theta}}) = \mathbf{x}_i$, and we obtain the FOC for the OLS estimator.

- We can directly show the conditional mean of the score is zero at θ_o :

$$\begin{aligned} E[\mathbf{s}(\mathbf{w}, \theta_o) | \mathbf{x}] &= -\nabla_{\theta} m(\mathbf{x}, \theta_o)' [E(y | \mathbf{x}) - m(\mathbf{x}, \theta_o)] \\ &= -\nabla_{\theta} m(\mathbf{x}, \theta_o)' \cdot 0 = \mathbf{0}. \end{aligned}$$

- By iterated expectations, $E[\mathbf{s}(\mathbf{w}, \theta_o)] = \mathbf{0}$.
- The Hessian is

$$\mathbf{H}(\mathbf{w}, \theta) = \nabla_{\theta} \mathbf{s}(\mathbf{w}, \theta) = \nabla_{\theta} m(\mathbf{x}, \theta)' \nabla_{\theta} m(\mathbf{x}, \theta) - \nabla_{\theta}^2 m(\mathbf{x}, \theta) u(\theta)$$

where $u(\theta) \equiv y - m(\mathbf{x}, \theta)$. Note that $\nabla_{\theta} m(\mathbf{x}, \theta)' \nabla_{\theta} m(\mathbf{x}, \theta)$ and $\nabla_{\theta}^2 m(\mathbf{x}, \theta)$ are $P \times P$.

- We can use $\mathbf{H}(\mathbf{w}, \theta)$ to estimate the asymptotic variance, or simply by using $E(u|\mathbf{x}) = 0$:

$$\begin{aligned}\mathbf{A}(\mathbf{x}, \theta_o) &= E[\mathbf{H}(\mathbf{w}, \theta_o)|\mathbf{x}] = \nabla_{\theta} m(\mathbf{x}, \theta_o)' \nabla_{\theta} m(\mathbf{x}, \theta_o) - \nabla_{\theta}^2 m(\mathbf{x}, \theta_o) E[u(\theta_o)|\mathbf{x}] \\ &= \nabla_{\theta} m(\mathbf{x}, \theta_o)' \nabla_{\theta} m(\mathbf{x}, \theta_o).\end{aligned}$$

- A common estimator of $\mathbf{A}_o$ for NLS uses $\mathbf{A}(\mathbf{x}, \theta_o)$:

$$\hat{\mathbf{A}} = N^{-1} \sum_{i=1}^N \nabla_{\theta} m(\mathbf{x}_i, \hat{\theta})' \nabla_{\theta} m(\mathbf{x}_i, \hat{\theta}) = N^{-1} \sum_{i=1}^N \nabla_{\theta} \hat{m}_i' \nabla_{\theta} \hat{m}_i,$$

but it assumes correct specification of the conditional mean.

- In linear case, this reduces to the usual estimator even if we are just estimating the linear projection.

- For $\mathbf{B}_o$, $\mathbf{s}_i = -\nabla_{\theta} m(\mathbf{x}_i, \hat{\theta})' [y_i - m(\mathbf{x}_i, \hat{\theta})] \equiv -\nabla_{\theta} \hat{m}_i' \hat{u}_i$ where

$$\hat{u}_i \equiv y_i - m(\mathbf{x}_i, \hat{\theta})$$

are the NLS residuals.

- Simple algebra gives

$$\hat{\mathbf{B}} = N^{-1} \sum_{i=1}^N \mathbf{s}_i' \mathbf{s}_i = N^{-1} \sum_{i=1}^N \hat{u}_i^2 \nabla_{\theta} \hat{m}_i' \nabla_{\theta} \hat{m}_i$$

- Combining with $\hat{\mathbf{A}}$ gives the Huber-White heteroskedasticity-robust variance matrix estimator:

$$\widehat{Avar}(\hat{\theta}) = \left(\sum_{i=1}^N \nabla_{\theta} \hat{m}_i' \nabla_{\theta} \hat{m}_i \right)^{-1} \left(\sum_{i=1}^N \hat{u}_i^2 \nabla_{\theta} \hat{m}_i' \nabla_{\theta} \hat{m}_i \right) \left(\sum_{i=1}^N \nabla_{\theta} \hat{m}_i' \nabla_{\theta} \hat{m}_i \right)^{-1}$$

(under correct specification of the conditional mean).

- The asymptotic variance simplifies under a homoskedasticity assumption.

ASSUMPTION NLS.3: $\text{Var}(y|\mathbf{x}) = \sigma_o^2$. $\square$

- With Assumption NLS.3, the expression for $\mathbf{B}_o$ simplifies, just as with OLS:

$$\begin{aligned} E[\mathbf{s}(\mathbf{w}, \theta_o)\mathbf{s}(\mathbf{w}, \theta_o)'|\mathbf{x}] &= E[u^2 \nabla_{\theta} m(\mathbf{x}, \theta_o)' \nabla_{\theta} m(\mathbf{x}, \theta_o) | \mathbf{x}] \\ &= E(u^2 | \mathbf{x}) \nabla_{\theta} m(\mathbf{x}, \theta_o)' \nabla_{\theta} m(\mathbf{x}, \theta_o) \\ &= \sigma_o^2 \nabla_{\theta} m(\mathbf{x}, \theta_o)' \nabla_{\theta} m(\mathbf{x}, \theta_o) \end{aligned}$$

because $E(u^2|\mathbf{x}) = \text{Var}(u|\mathbf{x}) = \text{Var}(y|\mathbf{x})$ when $E(u|\mathbf{x}) = 0$.

- Aside: NLS.3 does *not* say that $\text{Var}(y|\mathbf{x}) = \text{Var}(y)$.

- Under NLS.1 and NLS.3,

$$\begin{aligned}\mathbf{B}_o &= E[\mathbf{s}(\mathbf{w}, \theta_o)\mathbf{s}(\mathbf{w}, \theta_o)'] = \sigma_o^2 E[\nabla_{\theta} m(\mathbf{x}, \theta_o)' \nabla_{\theta} m(\mathbf{x}, \theta_o)] \\ &= \sigma_o^2 \mathbf{A}_o.\end{aligned}$$

- A consistent estimator of σ_o^2 (with a degree-of-freedom adjustment) is

$$\hat{\sigma}^2 = (N - P)^{-1} \sum_{i=1}^N \hat{u}_i^2$$

and then

$$\widehat{Avar}(\hat{\theta}) = \hat{\sigma}^2 \left(\sum_{i=1}^N \nabla_{\theta} \hat{m}_i' \nabla_{\theta} \hat{m}_i \right)^{-1}.$$

- Remember, the heteroskedasticity-robust formulation is valid with or without Assumption NLS.3.
- It *could* be the nonrobust form has better finite-sample properties under NLS.3 but there is no guarantee.
- Exact finite-sample analysis is not available. Simulations can be informative.

Large-Sample Inference

- The asymptotic standard errors of the $\hat{\theta}_j$ are obtained as the square roots of the diagonal elements of $\hat{\mathbf{A}}^{-1}\hat{\mathbf{B}}\hat{\mathbf{A}}^{-1}/N$ (or maybe a simplified form, if available).
- We have

$$\frac{(\hat{\theta}_j - \theta_{oj})}{se(\hat{\theta}_j)} \xrightarrow{d} Normal(0, 1).$$

- To test $H_0 : \theta_{oj} = a_j$, use an asymptotic t statistic:

$$t(\hat{\theta}_j, a_j) = \frac{(\hat{\theta}_j - a_j)}{se(\hat{\theta}_j)} \overset{a}{\sim} Normal(0, 1)$$

under the null hypothesis.

- Can obtain approximate confidence intervals, too: $\hat{\theta}_j \pm 1.96 \cdot se(\hat{\theta}_j)$ for a large-sample 95% CI.

- We can test multiple, nonlinear restrictions.
- Let H_0 be stated as

$$H_0 : \mathbf{c}(\boldsymbol{\theta}_o) = \mathbf{0}$$

where $\mathbf{c} : \Theta \rightarrow \mathbb{R}^Q$, so there are Q restrictions. Assume $\mathbf{c}(\cdot)$ is continuously differentiable on $\text{int}(\Theta)$ and that $\boldsymbol{\theta}_o \in \text{int}(\Theta)$, as before.

- Let $\mathbf{C}(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \mathbf{c}(\boldsymbol{\theta})$ be the $Q \times P$ Jacobian, and define $\mathbf{C}_o \equiv \mathbf{C}(\boldsymbol{\theta}_o)$.

- By the mean value theorem argument,

$$\begin{aligned}\sqrt{N}[\mathbf{c}(\hat{\boldsymbol{\theta}}) - \mathbf{c}(\boldsymbol{\theta}_o)] &= \nabla_{\boldsymbol{\theta}}\mathbf{c}(\boldsymbol{\theta}_o)\sqrt{N}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) + o_p(1) \\ &= \mathbf{C}_o\sqrt{N}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) + o_p(1).\end{aligned}$$

- So, under H_0 ,

$$\sqrt{N}\mathbf{c}(\hat{\boldsymbol{\theta}}) \xrightarrow{d} \text{Normal}(\mathbf{0}, \mathbf{C}_o\mathbf{A}_o^{-1}\mathbf{B}_o\mathbf{A}_o^{-1}\mathbf{C}_o')$$

$$\sqrt{N}\mathbf{c}(\hat{\boldsymbol{\theta}})'[\mathbf{C}_o\mathbf{A}_o^{-1}\mathbf{B}_o\mathbf{A}_o^{-1}\mathbf{C}_o']^{-1}\sqrt{N}\mathbf{c}(\hat{\boldsymbol{\theta}}) \xrightarrow{d} \chi_Q^2$$

- The *Wald statistic* is

$$\begin{aligned}W &= N\mathbf{c}(\hat{\boldsymbol{\theta}})'(\hat{\mathbf{C}}\hat{\mathbf{A}}^{-1}\hat{\mathbf{B}}\hat{\mathbf{A}}^{-1}\hat{\mathbf{C}}')^{-1}\mathbf{c}(\hat{\boldsymbol{\theta}}) \\ &= \mathbf{c}(\hat{\boldsymbol{\theta}})' \{\widehat{Avar}[\mathbf{c}(\hat{\boldsymbol{\theta}})]\}^{-1}\mathbf{c}(\hat{\boldsymbol{\theta}})\end{aligned}$$

where $\hat{\mathbf{C}} = \mathbf{C}(\hat{\boldsymbol{\theta}}) = \nabla_{\boldsymbol{\theta}}\mathbf{c}(\hat{\boldsymbol{\theta}})$.

- Under H_0 ,

$$W \xrightarrow{d} \chi_Q^2.$$

- The Wald statistic is convenient when the unrestricted model is easy to estimate. It is almost always available, and can be made robust by using the sandwich form of the asymptotic variance estimate.
- Typically, the Wald statistic is the default when reported by econometrics packages.

- The *score statistic* or *Lagrange multiplier statistic* is based only on the restricted estimate.
- Let $\tilde{\theta}$ be the estimator that solves

$$\min_{\theta \in \Theta} \sum_{i=1}^N q(\mathbf{w}_i, \theta)$$

subject to $\mathbf{c}(\theta) = \mathbf{0}$

- The score principle is to insert the restricted estimates into the unrestricted score, and then seeing “how far” the result is from zero.

- Based on a mean value expansion, can show under H_0 that

$$N^{-1/2} \sum_{i=1}^N \mathbf{s}(\mathbf{w}_i, \tilde{\boldsymbol{\theta}}) = N^{-1/2} \sum_{i=1}^N \mathbf{s}(\mathbf{w}_i, \boldsymbol{\theta}_o) + \mathbf{A}_o \sqrt{N}(\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) + o_p(1).$$

- By another mean value expansion,

$$\sqrt{N}\mathbf{c}(\tilde{\boldsymbol{\theta}}) = \sqrt{N}\mathbf{c}(\boldsymbol{\theta}_o) + \mathbf{C}_o \sqrt{N}(\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) + o_p(1).$$

- But $\mathbf{c}(\boldsymbol{\theta}_o) = \mathbf{0}$ under H_0 and $\mathbf{c}(\tilde{\boldsymbol{\theta}}) = \mathbf{0}$ because $\tilde{\boldsymbol{\theta}}$ is the restricted estimator. So $\mathbf{C}_o \sqrt{N}(\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) = o_p(1)$.

- Therefore, under H_0 ,

$$\mathbf{C}_o \mathbf{A}_o^{-1} \left(N^{-1/2} \sum_{i=1}^N \mathbf{s}(\mathbf{w}_i, \tilde{\theta}) \right) = \mathbf{C}_o \mathbf{A}_o^{-1} \left(N^{-1/2} \sum_{i=1}^N \mathbf{s}(\mathbf{w}_i, \theta_o) \right) + o_p(1)$$

so

$$\mathbf{C}_o \mathbf{A}_o^{-1} \left(N^{-1/2} \sum_{i=1}^N \mathbf{s}(\mathbf{w}_i, \tilde{\theta}) \right) \xrightarrow{d} \text{Normal}(\mathbf{0}, \mathbf{C}_o \mathbf{A}_o^{-1} \mathbf{B}_o \mathbf{A}_o^{-1} \mathbf{C}_o')$$

- The LM statistic is a quadratic form in $N^{-1/2} \sum_{i=1}^N \tilde{\mathbf{s}}_i$:

$$LM = \left(\sum_{i=1}^N \tilde{\mathbf{s}}_i \right)' \tilde{\mathbf{A}}^{-1} \tilde{\mathbf{C}}' (\tilde{\mathbf{C}} \tilde{\mathbf{A}}^{-1} \tilde{\mathbf{B}} \tilde{\mathbf{A}}^{-1} \tilde{\mathbf{C}}')^{-1} \tilde{\mathbf{C}}' \tilde{\mathbf{A}}^{-1} \left(\sum_{i=1}^N \tilde{\mathbf{s}}_i \right) / N \xrightarrow{d} \chi_Q^2.$$

- All quantities are evaluated at $\tilde{\boldsymbol{\theta}}$. For example,

$$\tilde{\mathbf{B}} = N^{-1} \sum_{i=1}^N \tilde{\mathbf{s}}_i \tilde{\mathbf{s}}_i'$$

- Depending on the choice of $\tilde{\mathbf{A}}$, it may not be positive definite (particularly the Hessian form). But the LM statistic above is always nonnegative.
- This is a fully robust LM statistic in that it only assumes θ_o minimizes $E[q(\mathbf{w}_i, \theta_o)]$ subject to $\mathbf{c}(\theta_o) = \mathbf{0}$ (and, the regularity conditions of θ_o being in the interior under H_0 , differentiability, and moment conditions).
- If the *generalized information matrix equality*, that is, for some $\sigma_o^2 > 0$,

$$\mathbf{B}_o = \sigma_o^2 \mathbf{A}_o$$

then the LM statistic simplifies to

$$LM = \left(\sum_{i=1}^N \tilde{\mathbf{s}}_i \right)' \tilde{\mathbf{M}}^{-1} \left(\sum_{i=1}^N \tilde{\mathbf{s}}_i \right) / \tilde{\sigma}^2,$$

where $\tilde{\sigma}^2$ is a consistent estimator of σ_o^2 .

- The matrix $\tilde{\mathbf{M}}$ can be one of the matrices

$$\sum_{i=1}^N \tilde{\mathbf{H}}_i, \quad \sum_{i=1}^N \tilde{\mathbf{A}}_i, \quad \sum_{i=1}^N \tilde{\mathbf{s}}_i \tilde{\mathbf{s}}_i'.$$

- These lead to the “Hessian,” “expected Hessian,” and “outer product” of the LM or score statistics. This is a nonrobust statistic because it uses the GIME.

- Under the GIME, a statistic based on the change in the criterion function is available.
- This requires estimation of both the restricted and unrestricted models but no matrix algebra for computation.
- We know that

$$\sum_{i=1}^N q(\mathbf{w}_i, \tilde{\theta}) \geq \sum_{i=1}^N q(\mathbf{w}_i, \hat{\theta})$$

because $\tilde{\theta}$ is the restricted estimator.

- Define the *quasi-likelihood ratio (QLR) statistic* as

$$QLR = 2 \left[\sum_{i=1}^N q(\mathbf{w}_i, \tilde{\boldsymbol{\theta}}) - \sum_{i=1}^N q(\mathbf{w}_i, \hat{\boldsymbol{\theta}}) \right] / \hat{\sigma}^2,$$

where $\hat{\sigma}^2$ is a consistent estimator of σ_o^2 typically obtained without the restrictions imposed.

- Under $H_0 : \mathbf{c}(\boldsymbol{\theta}_o) = \mathbf{0}$ and $\mathbf{B}_o = \sigma_o^2 \mathbf{A}_o$,

$$QLR \xrightarrow{d} \chi_Q^2$$

- The QLR name comes from its application to maximum likelihood estimation. But it is also related to the usual F statistic from linear regression.

- In fact, if $q(\mathbf{w}_i, \boldsymbol{\theta}) = (y_i - \mathbf{x}_i\boldsymbol{\theta})^2/2$ and $\hat{\sigma}^2 = (N - P)^{-1} \sum_{i=1}^N \hat{u}_i^2$, then $QLR = Q \cdot F$, where F is the usual F statistic for testing the Q restrictions.
- Generally, for NLS under NLS.3, we can use

$$F = \frac{(SSR_r - SSR_{ur})}{SSR_{ur}} \cdot \frac{(N - P)}{Q}$$

as approximately $\mathcal{F}_{Q, N-P}$ under H_0 .

- No theoretical reason not to use QLR as approximate χ_Q^2 , but the “ F ” statistic has been shown to sometimes have better size in not-so-large samples.

Choosing Among the Statistics

- Under the null, the robust versions of the Wald and LM statistics have the same limiting chi-square distribution. The QLR statistic does under the GIME, and, of course, there are nonrobust versions of the Wald and LM statistics.
- Can we use power considerations to choose among the statistics? Against fixed alternatives, all three statistics reject with probability approaching one. That is, they are *consistent tests*.
- Typically, the choice is based on computational simplicity and evidence of finite-sample performance.